LAKSHYA 2.0, LDCE

तर्क HACKATHON HAASTRA

DESIGN DECODE DOMINATE

Problem Statement Booklet

Tark Shaastra · Lakshya 2.0

24-Hour Hackathon · Team Size: 1–4 · Software Track

22 Problem Statements · 5 Tracks · Multiple Domains

Organised by: LDCE · Industry Partner: DEV IT

TRACK & DOMAIN OVERVIEW

Track	ID Range	Domains
Track 1 — AI / ML	TS-01, TS-02,TS-03, TS-04	Education · Public Grievances · Edu-Technical · Public Institution · Govt. Policy · Forest Security
Track 2 — Web3 / Blockchain	TS-05, TS-06, TS-07, TS-08	Edu-Technical · Govt. Policy · Forest Product Value Chain · Edu Innovation
Track 3 — Sustainable Tech	TS-09, TS-10	Environment · Water Conservation · Smart Plantation · Biodiversity
Track 4 — Smart Mobility & Infra	TS-11	Transportation · Tourism & Crowd Management
Track 5 — web-based monitoring	TS-12, TS-13, TS-14,	EDU INNOVATION

DATA SOURCE

Dear Participants,

For **Problem Statement 9**, it is **mandatory to use the dataset provided by us**.

Except Problem Statement 9 Please Note:

- The dataset shared is **synthetic data** and is intended specifically for your implementation.
- To use the dataset properly, **download the CSV file** from the drive and **open it using Excel (or any compatible tool)** before proceeding with your solution.

Kindly ensure you follow these instructions carefully while working on your submission.

https://drive.google.com/drive/folders/1Yvaxocp8f91-RKPBf6T97JJ77yjD1KpG?usp=sharing

PROBLEM STATEMENT INDEX

PS ID	Name	Domain
TS-01	Invisible Dropout Early Warning System	Education
TS-02	Grievance Resolution Verifier	Public Grievances
TS-03	Predictive Maintenance Engine — School Infrastructure	Public Institution
TS-04	DBT Leakage Detection System	Govt. Policy
TS-05	Verifiable Micro-Skill Passport for ITI & Polytechnic Graduates	Edu-Technical
TS-06	Scheme Eligibility on Chain — Tamper-Proof Entitlement Records	Govt. Policy
TS-07	Tamper-Proof Forest Product Chain-of-Custody System	Forest Product Value Chain
TS-08	Community Forest Rights Ledger & Benefit-Sharing Engine	Forest Product Value Chain
TS-09	Hyperlocal Pollution Evidence Engine	Environment
TS-10	Ghost Water Detector — Non-Revenue Water Localisation	Water Conservation
TS-11	Stampede Window Predictor for Gujarat Pilgrimage Corridors	Tourism & Crowd Management
TS-12	Early Academic Risk Detection & Student Intervention Platform	Web-based monitoring
TS-13	Intelligent Work Prioritization & Productivity Assistant	Web-based monitoring
TS-14	AI-Powered Complaint Classification & Resolution Recommendation Engine	Web-based monitoring

EDUCATION · AI / ML

TS-01
Invisible Dropout Early Warning System

PROBLEM STATEMENT

In Gujarat government schools, students rarely drop out suddenly — they drift away. Irregular attendance, declining scores, and household stress signals appear weeks before a formal dropout, yet no teacher has a structured, real-time decision tool to act. Build a system that ingests simple daily inputs — attendance, participation scores, mid-assessment marks — and produces a ranked at-risk list with student-specific action cards. Must function offline on a low-end Android tablet with 2G or offline sync.

WINNING LOGIC

- Risk score is explainable — teacher sees the reason, not just a number
- Action card is student-specific (e.g. 'Home visit — 11-day gap + failed Unit 2')
- Works offline / 2G on ₹8,000 Android device
- Teacher completes daily input in under 3 minutes
- 72-hour re-engagement SLA with principal escalation if unresolved

USE CASES

User Role	Key Use Case
Class Teacher	Mark daily attendance & scores; view ranked at-risk list; log re-engagement actions
School Principal	View school-level at-risk summary; assign re-engagement tasks; track success rate
DEO	Monitor district dashboard; receive escalation alerts; export compliance reports
Super Admin	Configure risk weightings; manage accounts; view state-wide analytics

FUNCTIONALITIES TO BE ACHIEVED

1. Daily input form: attendance + participation + assessment (< 3 min, offline-capable)
2. Multi-factor risk model: attendance gap + assessment decline + household stress
3. Explainable risk score with student-specific reason displayed to teacher
4. Student-specific action card with precise instruction and reason
5. 72-hour re-engagement SLA with automatic escalation to principal
6. Offline-first operation on ₹8,000 Android device; 2G sync
7. DEO district dashboard: school-wise at-risk heatmap and trend charts
8. Re-engagement outcome tracking — student retention recorded post-action

DATA SOURCE

Daily attendance registers, mid-assessment marks, participation scores submitted by class teachers. Historical DISE/U-DISE school records for baseline modelling.

PUBLIC GRIEVANCES · AI / ML

TS-02

Grievance Resolution Verifier

PROBLEM STATEMENT

Gujarat's Swagat portal marks thousands of complaints as 'resolved' monthly, yet a significant proportion receive only a text reply with no physical action taken. Build an AI-assisted verification layer that, when a department marks a grievance resolved, auto-triggers a geo-tagged photo upload from the field officer, a Gujarati IVR voice confirmation from the complainant, and a GPS-stamped field visit log. If verification fails, the grievance automatically reopens. Complainant loop works via basic phone — no smartphone required.

WINNING LOGIC

- Verification is independent of the resolving department — no self-certification
- Auto-reopen logic cannot be manually overridden by the same department
- Department quality score is public-facing and visible to the Collector in real time
- Complainant confirmation works via IVR — no smartphone or app needed
- Evidence packet (photo + GPS + voice) is archived and timestamped

USE CASES

User Role	Key Use Case
Citizen	Receives IVR call; confirms resolution via keypress; presses '2' to dispute and auto-reopen
Field Officer	Receives task; uploads geo-tagged photo + GPS-stamped visit log
Department Officer	Marks grievance resolved; system auto-triggers IVR + field evidence upload
District Collector	Views department quality scores; drills into failed verifications; escalates

FUNCTIONALITIES TO BE ACHIEVED

1. Auto-triggered IVR call to complainant on department resolution attempt
2. Gujarati-language IVR voice confirmation with keypress response
3. Geo-tagged photo upload from field officer as mandatory resolution evidence
4. GPS-stamped field visit log — location must match grievance address
5. Auto-reopen logic: grievance reopens if IVR confirmation absent or disputed
6. Department resolution-quality score visible to District Collector in real time
7. Verification independent of resolving department (no self-certification)
8. Jan Sunwai-ready audit packet: photo + GPS + voice log per grievance

DATA SOURCE

Swagat portal grievance records; field officer GPS logs; IVR call metadata.

PUBLIC INSTITUTION · AI / ML

TS-03

Predictive Maintenance Engine — School Infrastructure

PROBLEM STATEMENT

Gujarat has over 30,000 government school buildings, a large proportion of which have classrooms, toilets, or electrical fittings in disrepair. Repairs happen reactively — after failure is visible. Build a predictive maintenance system where school staff submit weekly structured condition inputs via a simple mobile form. The system aggregates inputs, learns deterioration patterns by building age, material, and weather zone, and generates a prioritised maintenance schedule for the DEO — flagging schools 30–60 days from critical failure across plumbing, electrical, and structural categories.

WINNING LOGIC

- Failure prediction is category-specific: plumbing, electrical, and structural tracked independently
- Prioritisation accounts for student impact — broken toilet in girls' school ranks above cracked storage wall
- Prediction cites specific inputs that triggered it — not a black-box score
- DEO receives a district-level maintenance queue, not school-by-school alert flood
- System learns from completed repairs — repair records update the deterioration model

USE CASES

User Role	Key Use Case
School Peon/Watchman	Opens weekly dropdown form; selects category and condition; uploads optional photo (< 2 min)
School Principal	Views school condition summary; approves urgent repair requests
DEO	Views prioritised district maintenance queue; assigns contractors; tracks SLA
Contractor	Receives work order; submits photo + GPS completion confirmation

FUNCTIONALITIES TO BE ACHIEVED

1. Structured weekly condition form: dropdown + photo, no free text — completable in < 2 minutes
2. Category-specific failure prediction: plumbing, electrical, structural tracked independently
3. Failure prediction with cited evidence — not a black-box score
4. Student impact prioritisation (girls' school toilet > cracked wall in storage room)
5. Predictive horizon: 30–60 day failure window per category per school
6. DEO district-level maintenance queue — not individual school alert flood
7. Contractor work order and completion confirmation workflow
8. Model learns from completed repairs — repair records update deterioration model

DATA SOURCE

Weekly condition reports from school staff; historical maintenance records; building age and material data from state education department.

GOVT. POLICY · AI / ML

TS-04
DBT Leakage Detection System

PROBLEM STATEMENT

Gujarat disburses welfare benefits through Direct Benefit Transfer across multiple schemes. Known failure modes include: funds reaching ineligible or deceased beneficiaries, middlemen accounts receiving transfers, funds credited but never withdrawn (beneficiary never notified), and duplicate payments under slightly different names. No real-time system flags these patterns before the next payment cycle. Build a transaction monitoring and anomaly detection system that ingests simulated DBT data across 3 schemes, flags specific leakage patterns, assigns a risk score to each flagged transaction, and generates a structured audit report for the District Finance Officer.

WINNING LOGIC

- Detects at least 4 distinct leakage patterns: deceased beneficiary, duplicate identity, undrawn funds, cross-scheme duplication
- Name matching handles Gujarati transliteration variations — not just exact string match
- Risk score per flagged transaction is explainable with specific evidence
- Audit report is structured for the DFO — prioritised queue, not a raw data dump
- System processes 10,000+ transactions in under 30 seconds

USE CASES

User Role	Key Use Case
District Finance Officer	Views prioritised investigation queue; assigns cases to field investigators; exports audit report
Scheme Verifier	Receives assigned case; conducts field visit; submits GPS-stamped verification result
Audit Team Member	Queries cross-scheme duplicate flags; generates compliance summary
State DBT Admin	Configures leakage pattern rules; views state-level risk heatmap

FUNCTIONALITIES TO BE ACHIEVED

1. Transaction monitoring across 3 schemes — ingests simulated DBT payment data
2. 4 leakage pattern detectors: deceased beneficiary, duplicate identity, undrawn funds, cross-scheme duplication
3. Gujarati transliteration-aware name matching (fuzzy — not exact string)
4. Risk score per flagged transaction with specific evidence citation
5. Prioritised investigation queue for DFO — not a raw data dump
6. Processing benchmark: 10,000+ transactions in under 30 seconds
7. Field investigation workflow with GPS-stamped verification submission
8. State-level risk heatmap by scheme, district, and leakage type

DATA SOURCE

Simulated DBT payment data across 3 welfare schemes; civil death register (for deceased detection); Aadhaar linkage records.

TS-05

Verifiable Micro-Skill Passport for ITI & Polytechnic Graduates

PROBLEM STATEMENT

An ITI graduate's paper marksheet says 'passed' but gives an MSME employer no way to verify which machines were actually operated or how practical competency compares to other applicants. Fake certificates are rampant; genuine ones are unverifiable. Build a blockchain-backed micro-skill passport where ITI trainers issue granular, assessed competency badges — not just course completion — that an MSME HR manager can verify instantly via QR code scan. No wallet or crypto knowledge required on the employer side. Trust model must be institution-independent.

WINNING LOGIC

- Employer verifies a credential in under 10 seconds — no account, wallet, or technical knowledge
- Credential is skill-level and machine-specific, not just 'course completed'
- Issuer is not required for verification — trustless architecture
- Revocation mechanism exists — fake or withdrawn credentials can be invalidated
- Trainer can issue credentials on a basic Android device with intermittent connectivity

USE CASES

User Role	Key Use Case
ITI Graduate	Receives QR-linked skill passport; shares QR with employers; views own credential history
ITI Trainer	Opens Android app; selects student + skill/machine; enters competency level; issues badge on blockchain
ITI Principal	Views all credentials issued; revokes a badge with reason logged on chain
MSME Employer	Scans QR → browser shows name, trade, skills, machines, competency levels, issuer, date — no account needed

FUNCTIONALITIES TO BE ACHIEVED

1. Blockchain-backed micro-skill passport — granular badges per skill/machine, not course completion
2. Offline credential issuance on basic Android — syncs to chain on connectivity
3. QR code verification — public, no wallet, no crypto knowledge, under 10 seconds
4. Trustless architecture — employer verification does not require contacting issuer
5. Revocation mechanism — withdrawn credentials flagged on-chain, visible on QR scan
6. Competency level system (1–4) per skill domain for each badge
7. Institution onboarding and trainer management by ITI Board Admin
8. State-wide credential issuance and verification analytics dashboard

DATA SOURCE

ITI student competency assessment records; national skill framework (trade → skill domain → competency levels); forest department permit registry.

GOVT. POLICY · WEB3 / BLOCKCHAIN

TS-06

Scheme Eligibility on Chain — Tamper-Proof Entitlement Records

PROBLEM STATEMENT

Gujarat's welfare schemes are subject to duplicate registrations, ghost beneficiaries, and eligibility data manipulation at the panchayat level. A BPL household may be entitled to multiple schemes but quietly disqualified by an inflated income figure in one portal, with no notification and no audit trail. Build a system where household eligibility assessment records are written to an immutable ledger. Changes trigger an on-chain event visible to the beneficiary and district auditor. A CSC operator can trace exactly which data point changed, when, by whom, and what scheme it affected. Beneficiary flagging must not require legal literacy.

WINNING LOGIC

- Every change to a household's eligibility data produces an on-chain event with actor, timestamp, and field changed
- Beneficiary can view their own record and flag a discrepancy in under 5 steps
- Auditor dashboard shows all data changes across a district — sortable by scheme, date, and panchayat
- System detects duplicate beneficiary registrations automatically
- No central admin can delete or silently override a record

USE CASES

User Role	Key Use Case
Beneficiary	Views own entitlement record in Gujarati; flags discrepancy in 5 steps with document upload
CSC Operator	Searches household by Aadhaar; updates a field — system records old/new value, operator ID, timestamp on-chain
District Audit Officer	Views all data changes across district; filters by scheme, date, panchayat; adjudicates disputes
Scheme Verifier	Submits field verification result with GPS stamp linked to household blockchain record

FUNCTIONALITIES TO BE ACHIEVED

1. Immutable eligibility ledger — every field change produces on-chain event (actor, timestamp, old/new value)
2. Duplicate beneficiary detection — automatic flag on Aadhaar/name/address match across schemes
3. Beneficiary self-view in Gujarati — own record + scheme eligibility status
4. Beneficiary dispute flagging — under 5 steps, no legal literacy required
5. District auditor dashboard — all data changes sortable by scheme, date, panchayat
6. On-chain change event notifications to beneficiary and auditor
7. No delete or silent override — corrections are append-only with full trail
8. Field verification submission with GPS stamp linked to household record

DATA SOURCE

Household eligibility data from state DBT portal; Aadhaar-linked beneficiary records; scheme eligibility criteria definitions.

FOREST PRODUCT VALUE CHAIN · WEB3 / BLOCKCHAIN

TS-07

Tamper-Proof Forest Product Chain-of-Custody System

PROBLEM STATEMENT

Gujarat's forest-based products — tendu leaves, timber, bamboo, medicinal herbs — pass through 4–6 hands between forest and consumer. At every handoff, mislabelling, adulteration, and substitution of illegally sourced material is possible and common. Build a chain-of-custody system where a forest product's journey is recorded on an immutable ledger from harvest: who harvested, from which licensed block, on which date, under which permit. Each handoff adds a verified transaction. The final buyer scans a QR code to see the complete provenance — each step verified by the relevant authority, not self-declared.

WINNING LOGIC

- Each supply chain step is verified by a distinct authority — not self-declared by the seller
- QR scan shows full provenance in plain language — no blockchain literacy required for the buyer
- Offline entry by harvester syncs when connectivity is available — no data loss
- System detects volume anomalies — if a depot ships more than it received, it is flagged
- Permit linkage is explicit — each harvest entry references a real forest department permit number

USE CASES

User Role	Key Use Case
Harvester	Opens offline mobile form; enters product, quantity, GPS, permit ID; syncs to chain on connectivity
Depot Manager	Records receipt quantity; creates dispatch transaction; system auto-flags if dispatch > received
Trader	Scans product QR; adds transport document hash; signs transaction; chain updated
End Buyer/Exporter	Scans product QR → views full chain: harvester, block, permit, date, all handoffs — in plain Gujarati/English

FUNCTIONALITIES TO BE ACHIEVED

1. Offline harvest entry form — syncs to blockchain on connectivity
2. Immutable chain-of-custody ledger (one record per handoff, verified by authority)
3. Volume anomaly detection — depot dispatch > received = auto-flag
4. Permit linkage: every harvest references a live forest department permit number
5. QR code generation per product batch — public read, no crypto knowledge
6. Full provenance view in plain Gujarati/English on QR scan
7. Wallet-less digital signature for each handoff participant
8. Cross-chain audit query: all records for a given permit

DATA SOURCE

Forest department harvest permits; product batch GPS records; depot receipt and dispatch logs; supply chain actor registry.

FOREST PRODUCT VALUE CHAIN · WEB3 / BLOCKCHAIN

TS-08

Community Forest Rights Ledger & Benefit-Sharing Engine

PROBLEM STATEMENT

Under the Forest Rights Act, tribal communities in Gujarat hold legal rights to harvest and benefit from specified forest produce. In practice, benefit-sharing between cooperatives, tribal families, and the forest department is opaque — payments are delayed, amounts disputed, and families have no record of what they are owed. Build a community forest rights ledger where each tribal family's entitlement is recorded on an immutable ledger. Each payment event is recorded with timestamp, amount, and authorising officer. Families check their entitlement and payment history via a QR-linked Gujarati interface. Disputes are logged with a structured evidence trail reviewable by the DTDO — without requiring family travel.

WINNING LOGIC

- Every payment is recorded with timestamp, amount, and authorising officer — not just a total balance
- Family can view their own entitlement and payment history without an intermediary
- Dispute is logged with structured evidence and routed to DTDO automatically
- System detects underpayment by comparing entitlement ledger to payment ledger
- Gujarati interface with voice support is functional for a low-literacy user

USE CASES

User Role	Key Use Case
Tribal Family	Scans QR card → views Gujarati interface: entitlement, payments received, shortfall; taps to log dispute with voice/photo
Cooperative Manager	Records payment event; system auto-checks entitlement ledger and flags underpayment
DTDO	Receives dispute notification with full evidence trail; adjudicates; marks resolution — no family travel required
Forest Dept. Officer	Issues/updates seasonal entitlement records; views underpayment reports

FUNCTIONALITIES TO BE ACHIEVED

1. Immutable entitlement ledger (produce type, quantity, value per season per family)
2. Payment event recording with timestamp, amount, authorising officer
3. Underpayment auto-detection: entitlement ledger vs. payment ledger
4. QR-linked family interface in Gujarati with voice support (low-literacy)
5. Dispute logging with voice/photo evidence — no office visit required
6. Structured evidence trail auto-routed to DTDO for adjudication
7. Dispute resolution workflow with status tracking
8. Cooperative-level payment compliance dashboard

DATA SOURCE

Seasonal entitlement records from forest department; cooperative payment logs; tribal household registry under Forest Rights Act.

ENVIRONMENT · SUSTAINABLE TECH

TS-09

Hyperlocal Pollution Evidence Engine

PROBLEM STATEMENT

Villages within 2 km of Gujarat's industrial corridors — Vapi, Ankleshwar, Vatva — regularly experience pollution surges that GSPCB's district-level monitoring stations never record. When a sarpanch complains, the response is: 'No reading on record.' Build a low-cost distributed sensing network architecture where simulated sensor nodes capture PM2.5, SO2, and wind data. A data fusion layer correlates the event with nearby industrial activity schedules and wind vectors to infer probable emission source direction. Output is not a chart — it is a GSPCB Form-A compliant complaint document that a sarpanch can generate in one step, with all required fields pre-filled.

WINNING LOGIC

- Output is a downloadable, pre-filled GSPCB Form-A complaint document — not just a graph
- Source direction inference uses wind vector + industrial schedule data
- Event detection distinguishes industrial surge from seasonal/agricultural sources
- Complaint generation takes under 60 seconds from event detection
- Sensor architecture is costed and justified under ₹2,500 per node

USE CASES

User Role	Key Use Case
Village Sarpanch	Views pollution surge alert; clicks 'Generate Complaint'; reviews pre-filled Form-A; submits to GSPCB in one step
GSPCB Inspector	Receives Form-A packet with all fields pre-filled; views source direction map; initiates inspection
Public Resident	Views village AQI on public dashboard; receives SMS/WhatsApp alert when PM2.5 exceeds threshold
GSPCB Admin	Manages sensor nodes; configures industrial schedule data; archives complaint documents

FUNCTIONALITIES TO BE ACHIEVED

1. Distributed sensor data ingestion: PM2.5, SO2, wind direction/speed (simulated nodes)
2. Industrial activity schedule correlation engine
3. Wind vector + industrial schedule → source direction inference
4. Industrial surge vs. agricultural/seasonal source classifier
5. GSPCB Form-A auto-generation with all required fields pre-filled from sensor data
6. One-step complaint submission by sarpanch (under 60 seconds from event detection)
7. Public village AQI dashboard (no login required)
8. Sensor node fault reporting and maintenance dispatch workflow

DATA SOURCE

Simulated PM2.5, SO2, CONTINUOUS AMBIENT AIR QUALITY; GSPCB industrial activity schedules; GSPCB Form-A field definitions.

WATER CONSERVATION · SUSTAINABLE TECH

TS-10

Ghost Water Detector — Non-Revenue Water Localisation

PROBLEM STATEMENT

Gujarat's municipal water utilities lose 35–55% of treated water to non-revenue water (NRW) — leaks, illegal connections, and meter tampering — before it reaches a billed tap. What no utility can answer in real time is: where exactly is the loss, and what type is it? Build a pressure-pattern analysis system using simulated time-series data from a 3-zone distribution network. The system detects anomaly signatures indicating specific NRW types, localises the event to a 50-metre pipe segment, and dispatches a prioritised, structured work order to the nearest field crew via WhatsApp message — no new app required.

WINNING LOGIC

- Anomaly is classified by type — not just flagged as 'unusual'
- Pipe segment localisation is shown on a real map layer (AMC ward boundary as base)
- Field crew dispatch message is WhatsApp/SMS formatted — no app needed
- System quantifies estimated daily water volume loss per detected anomaly
- Morning demand peak is correctly excluded from false positives

USE CASES

User Role	Key Use Case
Zone Network Engineer	Receives anomaly alert: pipe segment, NRW type, map location, urgency; dispatches field crew via WhatsApp
Field Maintenance Crew	Receives WhatsApp message with location pin, anomaly type, urgency; travels to site; replies 'DONE' to confirm
Utility Manager	Views city-wide dashboard: active anomalies map, daily volume loss (litres), zone-wise NRW %
Data Analyst	Queries anomaly history by zone/date/type; exports CSV; validates false positive rate

FUNCTIONALITIES TO BE ACHIEVED

1. Pressure time-series anomaly detection on simulated 3-zone distribution network
2. NRW type classifier: pipe burst vs. slow seepage vs. illegal tap
3. 50-metre pipe segment localisation on map layer (AMC ward boundary base)
4. Morning demand peak exclusion — false positive prevention
5. WhatsApp/SMS work order dispatch — no new app required for field crew
6. Work order includes: segment location pin, anomaly type, urgency level
7. Daily estimated water volume loss quantification per anomaly

DATA SOURCE

Simulated time-series pressure sensor data from 3-zone municipal distribution network; AMC/JMC ward boundary GIS layer; historical demand pattern data.

TOURISM & CROWD MANAGEMENT · SMART MOBILITY & INFRA

TS-11

Stampede Window Predictor for Gujarat Pilgrimage Corridors

PROBLEM STATEMENT

Ambaji, Dwarka, Somnath, and Pavagadh collectively receive over 1.2 crore pilgrims during Navratri. Stampede risk forms 200–500 metres upstream in narrow access corridors and choke alleys — not at the temple gate. CCTV exists but is watched reactively. Build a real-time crowd pressure intelligence system using simulated corridor entry flow rates, transport arrival bursts, and corridor width constraints. The system computes a corridor pressure index, predicts a crush-risk window 8–12 minutes ahead, and triggers a coordinated alert to police, temple trust, and transport authority on a single shared dashboard — not separate WhatsApp groups.

WINNING LOGIC

- Corridor pressure index predicts risk 8–12 minutes before threshold breach
- All 3 agencies receive coordinated alert on one shared dashboard — not separate channels
- Alert-to-acknowledgement time is measured and displayed per event
- System includes a replay mode to demonstrate logic on a simulated near-crush scenario
- Correctly distinguishes genuine crush buildup from a momentary surge that self-resolves

USE CASES

User Role	Key Use Case
All 3 Agencies	View shared dashboard: pressure index, predicted crush window, recommended action per agency; acknowledge within 90 seconds
District Police	Receives crush-risk alert; deploys officers to choke point; logs response time
Temple Trust	Activates darshan hold at inner gate; redirects overflow queue; confirms action
GSRTC Transport Control	Holds incoming vehicles at 3 km holding point; logs hold action and expected pressure reduction

FUNCTIONALITIES TO BE ACHIEVED

1. Corridor pressure index computation from entry flow rates + transport arrival bursts + width constraints
2. Crush-risk window prediction 8–12 minutes ahead of threshold breach
3. Genuine crush buildup vs. momentary self-resolving surge classifier
4. Single shared dashboard for all 3 agencies — not separate WhatsApp groups
5. Alert-to-acknowledgement response time tracked and displayed per event
6. Coordinated alert with agency-specific recommended action
7. Replay mode: demonstrate prediction logic on simulated near-crush scenario
8. Event log archive with pressure timeline and response timeline

DATA SOURCE

Simulated corridor entry flow rates; GSRTC vehicle arrival schedule data; historical pilgrimage crowd count data; corridor widthand chokepoint GIS data.

EDU INNOVATION · WEB-BASED MONITORING

TS-12
Early Academic Risk Detection & Student Intervention Platform

Dev IT Limited

PROBLEM STATEMENT

In universities and colleges, academically at-risk students are often identified too late. Key indicators such as declining attendance, low internal marks, missed assignments, and irregular LMS activity are scattered across multiple systems and reviewed manually.

Faculty mentors rely on delayed reports or subjective judgment, resulting in late interventions such as counselling or remedial classes. By the time risk is identified, student performance and morale have already declined.

A unified system is needed to continuously analyze academic signals, detect early risk trends, and provide clear reasons for intervention—enabling proactive academic support and improved student outcomes.

WINNING LOGIC

Multi-factor risk scoring: System must calculate student risk using multiple academic indicators such as attendance percentage, internal marks, and assignment completion status

Explainable insights: Each at-risk student must have clearly displayed top contributing reasons (e.g., low attendance, missing assignments) instead of just a score

Actionable faculty dashboard: Faculty should be able to filter and view at-risk students by class, subject, and severity level for quick decision-making

Intervention recording: Mentors must be able to log specific actions such as counselling sessions, remedial classes, or assignment extensions

Performance improvement tracking: System should show comparison of student performance before and after intervention to measure effectiveness

Timely alerts: Automated alerts should notify mentors about high-risk students early (before major exams)

USE CASES

Student	View personal academic risk score with reasons; receive improvement suggestions such as increasing attendance or completing pending assignments; track progress over time after interventions
Faculty Mentor	Identify at-risk students from dashboard; analyze reasons behind risk; conduct counselling or academic support sessions; record intervention details and monitor student improvement
Subject Teacher	Upload internal assessment marks and assignment data; monitor subject-wise

	performance trends; identify students struggling in specific subjects
Academic Coordinator	View department or institution-level risk analytics; monitor effectiveness of interventions; identify patterns.

FUNCTIONALITIES TO BE ACHIEVED:

1. Academic risk score generation per student
2. Explanation engine showing contributing academic factors
3. Faculty dashboard with filters by class, subject, and risk level
4. Student self-view portal with progress indicators
5. Intervention logging system (mentor actions + remarks)
6. Pre- and post-intervention performance comparison
7. Automated alerts for high-risk students
8. Downloadable academic risk summary reports (PDF/CSV)

DATA SOURCE

College event and project submission forms; GTU affiliated college registry (400+ institutions); student participation records.

TS-13

Intelligent Work Prioritization & Productivity Assistant

Dev IT Limited

PROBLEM STATEMENT:

In fast-paced teams, task prioritization is often handled manually, leading to inefficient execution. High-impact or urgent tasks are sometimes overlooked, while less important tasks consume valuable time.

Managers lack real-time visibility into whether teams are focusing on the most critical tasks aligned with business goals. Since priorities are static and not dynamically updated, teams struggle to adapt to changing deadlines, workloads, and project requirements.

A system is required that intelligently analyzes task parameters and continuously updates priorities in real-time, ensuring optimal productivity and better alignment with organizational objectives.

WINNING LOGIC

- **Multi-factor priority scoring:** Task priority must be calculated using multiple inputs such as deadline, effort required, and business impact rather than fixed rules
- **Real-time dynamic prioritization:** Task order should automatically update whenever there are changes in deadlines, task status, or workload conditions
- **Actionable task sequencing:** System should not just assign priority labels but also suggest an optimized order of execution for users
- **Manager visibility dashboard:** A clear dashboard must show team workload distribution, task priorities, and bottlenecks for better decision-making
- **Adaptive learning from completion:** System should track completed tasks and adjust future prioritization logic based on past performance and patterns
- **Timely alerts and reminders:** Notifications should be generated for overdue tasks and approaching deadlines to ensure timely completion

USE CASES

Team Member	Create and update tasks with details like deadline and effort; view dynamically prioritized task list; follow suggested execution order; mark tasks as completed and track personal productivity
Faculty Mentor	Monitor team tasks and workload distribution; review and adjust task inputs such as deadlines

	or priorities; ensure high-impact tasks are being handled first; track overall project progress
Subject Teacher	View organization-level dashboards; analyze productivity trends across teams; identify inefficiencies or delays; generate reports for performance evaluation and decision-making

FUNCTIONALITIES TO BE ACHIEVED:

1. Task creation with attributes (title, deadline, effort, impact)
2. AI-based priority scoring system
3. Dynamic task reordering based on real-time updates
4. Daily prioritized task view for users
5. Alerts for overdue tasks and approaching deadlines
6. Manager dashboard for workload and priority visibility
7. Task completion tracking system
8. Reporting and export functionality (PDF/CSV)

DATA SOURCE

College event and project submission forms; GTU affiliated college registry (400+ institutions); student participation records.

TS-14
AI-Powered Complaint Classification & Resolution Recommendation Engine

Dev IT Limited

PROBLEM STATEMENT:

In the wellness business, customer complaints are received through multiple channels such as call centres, emails, and direct communication. These complaints are manually reviewed and categorized (e.g., product issues, packaging issues, trade inquiries), leading to delays, inconsistent tagging, and missed service-level agreements (SLAs).

There is no intelligent mechanism to prioritize critical complaints or suggest appropriate resolution steps. As a result, response time increases and customer satisfaction declines.

A system is needed that can automatically classify complaints, assign priority based on urgency, and recommend actionable resolution steps in real-time—ensuring faster, consistent, and efficient customer support operations.

WINNING LOGIC:

- **Intelligent complaint classification:** System must classify complaints using meaningful logic (combination of keyword analysis and AI/ML models), not static rule-based mapping
- **Dynamic priority tagging:** Priority levels (High/Medium/Low) should be assigned based on complaint content, urgency, and impact rather than manual input
- **Actionable resolution recommendations:** System must suggest specific next steps (e.g., replacement, escalation, follow-up) instead of generic responses
- **Real-time processing:** Classification, priority tagging, and recommendation must be generated within seconds of complaint submission
- **Comprehensive dashboard visibility:** Dashboard should provide real-time insights into complaint categories, priorities, and workload distribution
- **Lifecycle tracking**: System must track complaints from submission to resolution, including status updates and SLA compliance

USE CASES

Customer Support Executive	Submit complaints via text/email/call summary; view auto-classified complaint category and assigned priority; follow system-recommended resolution steps; update complaint status after action
Quality Assurance Team	Analyze categorized complaints for accuracy; monitor trends in complaint types; identify

	recurring issues; ensure consistency in classification and resolution quality
Operations Manager	Monitor real-time dashboards showing complaint distribution and priorities; track SLA compliance and resolution time

FUNCTIONALITIES TO BE ACHIEVED:

1. Complaint input system (text/email/call summaries)
2. Automatic complaint classification (Product / Packaging / Trade)
3. Priority tagging (High / Medium / Low)
4. AI-based resolution recommendation engine
5. SLA tracking for each complaint
6. Dashboard with complaint category and priority insights
7. Complaint history and status tracking
8. Reporting and export functionality (PDF/CSV)

DATA SOURCE

College event and project submission forms; GTU affiliated college registry (400+ institutions); student participation records.

EVALUATION CRITERIA

All teams are evaluated on the same universal rubric by the judging panel.

CRITERIA	WEIGHT
Innovation & Creativity	**25 marks**
Technical Complexity & Implementation	**25 marks**
Problem Relevance & Impact	**20 marks**
Demo & Presentation Quality	**15 marks**
Code Quality & Documentation	**10 marks**
Checkpoint Progress Score	**5 marks**
TOTAL	**100 marks**